

Open-source finite-element tests of the Rosenzweig–Porter hypothesis for freely vibrating plates: preliminary results

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July 4, 2026 — draft v0.2 (companion to the “Why Rosenzweig–Porter?” program)

Abstract

The companion paper *Why Rosenzweig–Porter?* proposes that the intermediate (Rosenzweig–Porter, RP) spectral statistics observed in freely vibrating thin plates arise from evanescent-wave coupling at free edges acting on an integrable base, and it registers the numerical tests that would decide the hypothesis, assigning them to COMSOL “or finite elements”. We report the first execution of that program with a fully open-source stack (scikit-fem, Argyris elements, certified eigensolvers), under preregistered readings and measurement-based accuracy gates. Main findings: (i) the boundary-controlled Poisson→intermediate transition is confirmed at 3.6σ ; (ii) the geometry program yields a three-tier hierarchy governed by the boundary’s adaptedness to separable coordinates — exact reduction (disk) gives Poisson; coordinate-adapted boundaries (ellipse, sector, rectangle) give weak sparse coupling; unadapted boundaries (triangle, superellipse $p \neq 2$) give full coupling — with a $+12.2\sigma$ step between the ellipse ($p = 2$) and the smooth soft-cornered superellipse ($p = 3$); (iii) Gap A, executed for the first time on the true operator, shows N -independent inverse participation ratios: at accessible sector sizes ($N \leq 324$) each free-plate mode is ≈ 1 – 2 free-free beam products, i.e. sparse avoided-crossing hybridization rather than an RP-fractal phase. The results support the free-edge coupling mechanism while sharpening it in two ways that only true-operator computations could reveal. All experiments, gates, and raw data are reproducible from the repository.

1 Introduction

López-González and coworkers reported that flexural spectra of freely vibrating rectangular plates exhibit avoided crossings and intermediate spacing statistics within each symmetry sector, well fitted by the Rosenzweig–Porter (RP) random-matrix model, while simply supported plates are Poissonian [1]; the finding extends to a freely vibrating disk sector [2]. The companion hypothesis paper [3] formalizes a mechanism — evanescent components of free-edge eigenfunctions generate an inter-mode coupling V on an integrable base H_0 — and registers the calculations that would support or challenge it. This note reports the first systematic execution of those calculations, replacing COMSOL with an open-source pipeline so that every number can be re-derived from a commodity workstation.

Throughout, $D = \rho h = 1$, eigenvalues $\Lambda = \omega^2$ of $D\nabla^4 w = \rho h \omega^2 w$, and the canonical rectangle has $a = 1$, $b = 1/1.6189043236$, $\nu = 0.33$. Spacing-ratio statistics $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$ are used without unfolding (Poisson 0.3863, GOE 0.5307 [4]), always per symmetry sector, always against *finite-size baselines computed with the identical protocol* (never asymptotic constants); the lowest ten levels per sector are dropped.

2 Methods and certification

Stack. Kirchhoff–Love bending energy discretized with quintic C^1 Argyris triangles (scikit-fem); free edges are the natural boundary conditions (no constraints), simply supported edges are reached through a validated Winkler penalty $V_n + \kappa w = 0$ with $\kappa = 10^{10}$ ($O(1/\kappa)$ error, 3×10^{-6} measured against the exact SSSS spectrum). Eigenvalues of record come from shift-invert Lanczos without eigenvectors; eigenvectors from banded shift-invert subspace iteration with deflation and whitened Rayleigh–Ritz projections, certified per mode by Rayleigh-quotient matching against the eigenvalues of record and residual gates. Symmetry labels come from mirror/rotation-character classification with simultaneous diagonalization of reflection Grams inside near-degenerate clusters ($\mathbb{Z}_2 \times \mathbb{Z}_2$, C_{3v} including exact E doublets, and single-reflection variants).

Accuracy policy. Statistics-grade means eigenvalue error below $0.1 \times$ the local mean level spacing for every mode used. Every experiment runs behind gates: an exact or semi-analytic anchor where one exists (SSSS rectangle; Lamé² for the simply supported triangle; the per- m Bessel J/I free-disk determinant, fundamental $x^2 = 5.262$ vs. literature ≈ 5.25), two-mesh internal checks otherwise, and — for curved boundaries, where inscribed-polygon geometry error caps convergence at $O(h^2)$ — an *operational* gate: the FEM disk’s class-sequence statistics must reproduce the semi-analytic Poisson control (0.0σ agreement measured over 1600 modes; the non-smooth error component is bounded at 1.4×10^{-5} relative, $\sim 0.3\%$ of a sector spacing). Readings were preregistered in each experiment’s README and committed before execution.

3 Results

3.1 E2: boundary-controlled transition (supports, 3.6σ)

Sweeping the symmetric Winkler restraint from free ($\kappa = 0$) to simply supported ($\kappa \rightarrow \infty$) on the rectangle, the pooled per-sector $\langle r \rangle$ interpolates monotonically from 0.4422 ± 0.0096 to 0.3908 ± 0.0104 at ~ 200 levels/sector. Both endpoints match their finite-size baselines exactly: the SS end equals the exact-SSSS same-window value to four decimals, and the free end matches the independent Legendre–Ritz reference. The transition amplitude is 3.6σ above the separable baseline (preregistered threshold 3σ); no intermediate statistics appear at nearly-simply-supported coupling (Fig. 1).

3.2 E3: geometry program

Triangle (supports, 6.8σ). The free equilateral triangle — Helmholtz-integrable, biharmonically non-separable, symmetry group C_{3v} with exact E doublets — gives pooled $\langle r \rangle = 0.4889 \pm 0.0099$ vs. a same-protocol simply supported baseline 0.3753 ± 0.0134 (the SS polygon plate factorizes through the Dirichlet Helmholtz problem, i.e. the Lamé lattice squared, matched by the FEM to 3×10^{-5} across all 1000 modes). All three sectors separate positively ($A_1 + 3.8\sigma$, $E + 5.6\sigma$, $A_2 + 1.9\sigma$); 333 E doublets split at $< 10^{-5}$ (Fig. 2).

Disk control (confirmed) and ellipse. The full free disk — free edges with preserved separability — is the program’s zero-coupling control: the semi-analytic class sequence gives $\langle r \rangle = 0.3905 \pm 0.0073$ (the companion program’s executed control: 0.386 ± 0.007), and each fixed- m sequence is a picket fence. The free ellipse ($a/b = 1.6189$, area π), predicted intermediate under the

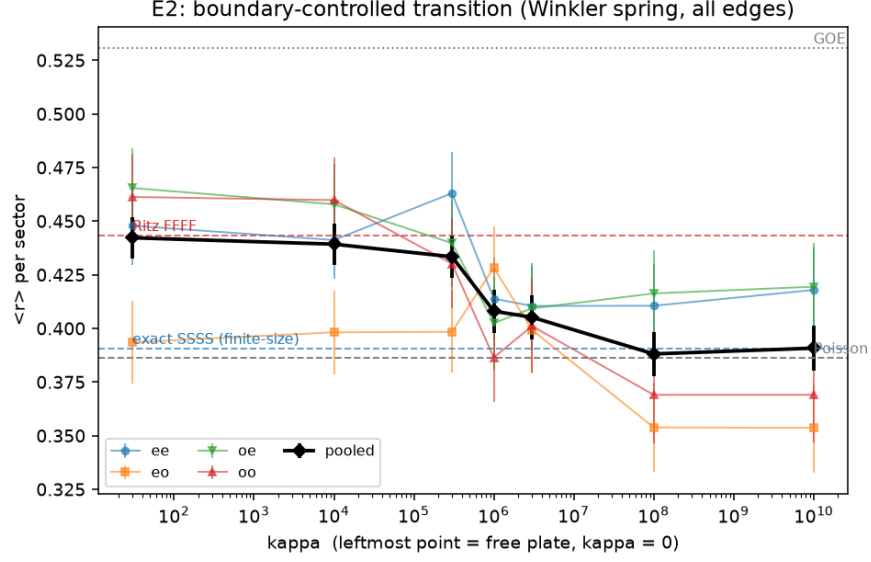


Figure 1: E2: per-sector and pooled $\langle r \rangle$ vs. the edge-spring constant κ (800 modes, Argyris 128×80).

naive reading, is Poisson-consistent at 1600 modes (0.3732 ± 0.0071); a suggestive upward trend at 800 modes did not survive the registered long-ladder decider (Fig. 3).

Disk sector (ambiguous, $+1.7\sigma$). At $\theta = 2.0$ rad and 800 modes the sector's two reflection classes sit $+1.2\sigma/+1.7\sigma$ above the disk baseline (pooled 0.4213 ± 0.0095) — the same tendency as the published COMSOL sector study but below our preregistered threshold; a decisive comparison needs their mode counts ($\sim 10^4$) or exact angles (Fig. 4).

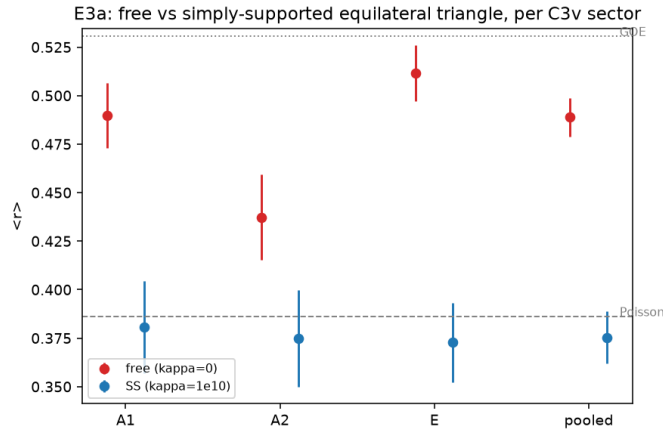


Figure 2: E3a: free vs. simply supported equilateral triangle per C_{3v} sector.

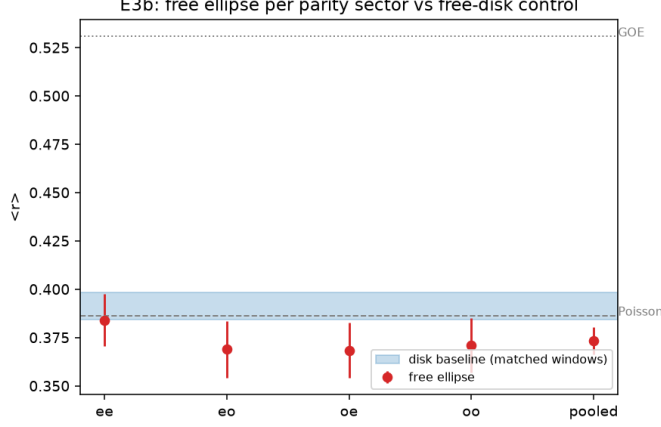


Figure 3: E3b (v2 decider): free-ellipse parity sectors vs. the free-disk control at 1600 modes.

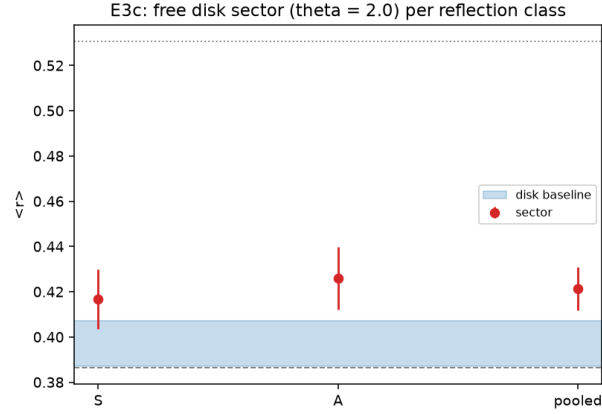


Figure 4: E3c: free disk sector ($\theta = 2$) per reflection class.

3.3 E5: superellipse sweep (the discriminator)

The family $|x/a|^p + |y/b|^p = 1$ breaks biharmonic separability equally for every p while corner-region curvature grows with p at fixed C^∞ smoothness. The measured sequence,

$$\langle r \rangle(p) = 0.3732, 0.4944, 0.4893, 0.4942, 0.5159 \quad (p = 2, 3, 4, 6, 10),$$

is a $+12.2\sigma$ step between $p = 2$ and $p = 3$, a flat plateau through $p = 6$, and a mild $+2.3\sigma$ rise at $p = 10$ (Fig. 5); the statistics are refinement-stable at 0.1σ for both the headline and worst-corner points. Neither preregistered shape (curvature-graded; true-corner threshold) fits: the operative variable is the boundary's compatibility with separable coordinates — the ellipse is the family's unique quasi-separable member (confocal boundary = elliptic-coordinate line) — and corners are secondary.

3.4 E4: Gap A on the true operator (localized-leaning at $N \leq 324$)

The companion paper's decisive test: eigenvector statistics per sector, in the two registered representation bases (simply supported sine products and free-free beam products), against empirical

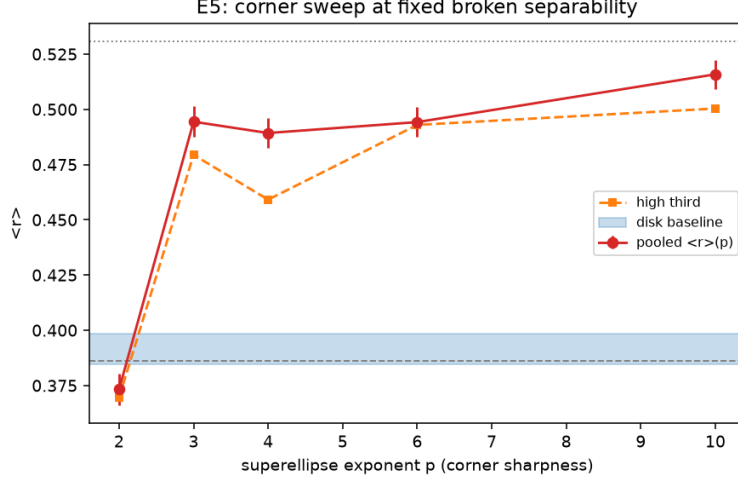


Figure 5: E5: $\langle r \rangle(p)$ along the superellipse family at 1600 modes per geometry.

GOE baselines at identical N (the T1 audit protocol). On 1200 certified eigenpairs (vector residuals $\leq 8 \times 10^{-6}$), the mean window IPR is *independent of* N across the ladder $N = 64$ –324 in both bases (ladder $D_2 \approx 0.02$ –0.13; GOE ≈ 1), and in the beam basis — which represents the true modes with captured norm 0.999–1.000 — each mode is ≈ 1 –2 beam products (IPR ≈ 0.7 ; Fig. 6). Per the preregistered dichotomy this is the localized/banded side at accessible N : sparse strong pair-hybridizations (microscopically, the observed avoided crossings) produce intermediate spacing statistics without RP-fractal eigenvectors. Two sharpenings: the true-operator D_2 is *lower* than the Ritz-model values (beam fixed- N 0.25 vs. 0.35–0.42), i.e. the model route overestimates coupling density; and since the effective coupling grows with mode index (E2, E3b), the verdict could still flip at the registered $N \sim 2048$ rung, reachable via quarter-plate reduction. $\Sigma^2(L \leq 5)$ is intermediate, mildly below Poisson; larger L is unfolding-limited.

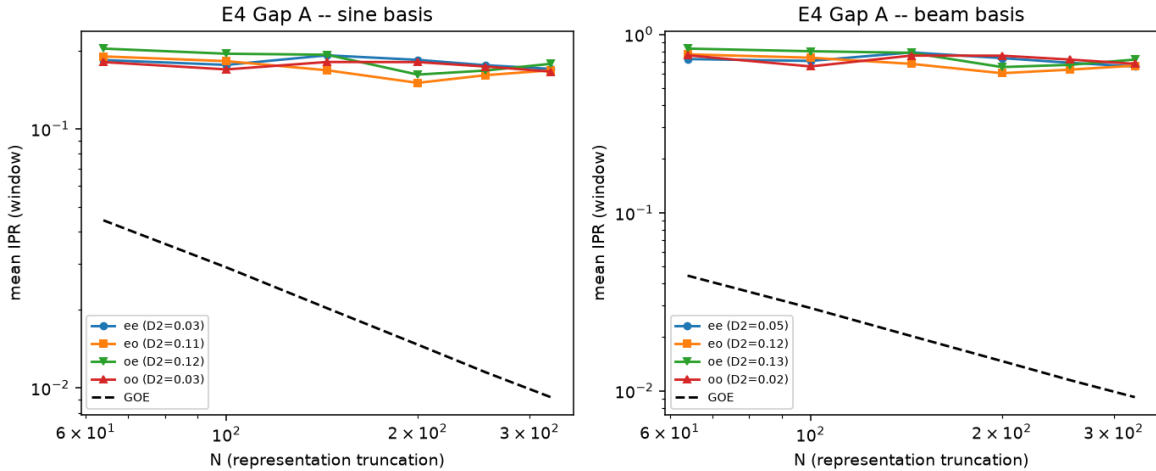


Figure 6: E4: window-mean IPR vs. representation truncation N , per sector, in the sine (left) and beam (right) registered bases, with GOE baselines.

3.5 E6: designed contact coupling and the control law

The companion’s contact test, executed with certified mode shapes at contact points on two identical coupled plates (exact reduction: symmetric combinations lie in the coupling’s kernel; the antisymmetric block feels $H_0 + 2\lambda C$ with $C_{ij} = q \phi_i(x_k) \phi_j(x_k)$). The *eigenvector* dichotomy of the prediction is crisp — dense-pattern mid-spectrum IPR 0.033 (extended) vs. strip-confined 0.18–0.19 (localized), central post rank-1 as designed — while the spacing dichotomy points the right way at 2.3σ (both patterns rank-deficient at 120 quartets; a full-rank rerun is registered). The P10 *control law* — post position from the nodal maps, stiffness $\lambda = \Delta/(2q\phi_i^2(x^*))$ plus a closed-form second-order correction — reproduces the designed splitting to 3.6% at $\Delta = 0.25 d_{\text{loc}}$ (first order alone: 19.7%), degrading exactly as the target pair dissolves into its neighbors; the validity boundary ($\Delta \lesssim 0.5 d_{\text{loc}}$) is itself predicted by the pair-content collapse.

3.6 E7: Lévy configuration and material tunability

Protocol A’s mixed configuration (free x -edges, simply supported y -edges) is the classical Lévy-separable plate; the hierarchy therefore requires Poisson, and the measurement lands on the baseline (0.3973 ± 0.0103): one free edge pair does not suffice — intermediate statistics require free edges that break *every* product structure. The ν -sweep reproduces the companion repository’s Ritz values value-by-value (≈ 0.003 agreement between independent methods): $\langle r \rangle$ rises monotonically $0.4208 \rightarrow 0.4497$ over $\nu = 0.25\text{--}0.36$, so the effective coupling is *material-tunable* (the Prediction-5 recast) rather than strictly material-independent.

3.7 E8: the eigenvector hierarchy across geometries

Repeating Gap A with the same-domain simply supported eigenbasis (exact M -inner-product coefficients) for four geometries (Fig. 7): the eigenvector structure *mirrors* the E5 spacing hierarchy. Adapted boundaries have flat-IPR sparse eigenvectors — rectangle ≈ 0.18 ($D_2 \approx 0$, cross-validating E4 method-to-method) and ellipse ≈ 0.40 : the registered ellipse decider lands between its poles, *tiny sparse coupling* (~ 2.5 SS modes per free mode, N -independent). Unadapted boundaries have genuinely *scaling* eigenvectors — triangle $D_2 = 0.27/0.36$, superellipse $p=10$ $D_2 = 0.22\text{--}0.50$ — the RP-candidate regime, sitting exactly where the spacing statistics are strongest. The registered RP-vs-PBRM D_q separator splits at these N (superellipse flat- D_q /RP-leaning, triangle multifractal-leaning; both D_2 well below the P12 reference 0.76 ± 0.15), so that adjudication is registered for the $N \sim 2048$ rung.

4 Synthesis

Seven geometries fall on one hierarchy:

tier	geometries ($\langle r \rangle$ at matched ladders)	mechanism state
exact reduction	disk 0.391	no coupling (Poisson)
coordinate-adapted	ellipse 0.373; sector 0.421; rectangle 0.44	weak, sparse coupling
unadapted	triangle 0.489; superellipse $p \neq 2$: 0.49–0.52	full coupling

The free-edge origin of the coupling is confirmed directly (E2), the Lévy configuration (E7) adds an exactly-reduced member to the top tier, and the eigenvector program (E4, E8) shows the hierarchy is not only spectral: adapted boundaries carry sparse, N -independent eigenvector spreads (rectangle

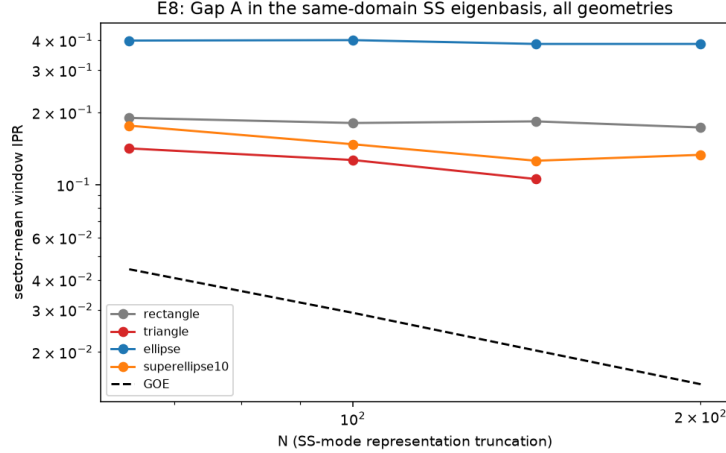


Figure 7: E8: sector-mean window IPR vs. representation truncation in the same-domain SS eigenbasis, all geometries, with GOE baselines.

$\approx$ beam products; ellipse ≈ 2.5 SS modes), while unadapted boundaries carry genuinely scaling eigenvectors ($D_2 \approx 0.2\text{--}0.5$) — the RP-candidate regime living exactly where the spacing statistics are strongest. What the data refine is the *selector*: not corners, not curvature, but whether the boundary is adapted to a separable structure of the fourth-order operator. The effective coupling is also material-tunable ($\lambda(\nu)$, E7) and post-designable with a quantitative control law (E6) — both usable knobs for the experimental program. Whether the scaling eigenvectors of the unadapted tier become RP-fractal (flat D_q) or PBRM-multifractal at higher rungs is the single most consequential open computation.

5 In execution and pending registered tests

In execution at the time of this draft revision: **E10** (annulus, the second exactly-reduced Poisson control, via a 4×4 $J/Y/I/K$ free-edge determinant with FEM cross-validation) and **E11** (Mindlin–Reissner thickness sweep; formulation validated against the certified Kirchhoff spectrum in the thin limit). Pending, registered in the research plan: Gap A at $N \sim 2048$ via a C^0 interior-penalty discretization (the Argyris uniform-mesh ceiling is measured at $h \approx 0.008$) — this adjudicates both the RP-phase question and the D_q separator; the sector at the published study’s exact angles and mode counts; a full-rank rerun of the E6 dichotomy; and comparison against the measured aluminum-plate spectra of Ref. [1].

6 Reproducibility

All experiments live in `experiments/eNN_*` of the private repository `plates-rp-fem` with frozen code, preregistered README, RESULTS, raw JSON, and figures; the shared machinery is the `platefem` package (conda env `plates-fem`: Python 3.12, scikit-fem 12.0.2, SciPy 1.18). Total compute for E1–E5: under 8 hours on one 8-core desktop (Ryzen 7 5800XT, 64 GB). Two numerical contributions may be of independent use: a certified eigenpair pipeline that works around a SciPy 1.18 shift-invert eigenvector-path failure on ill-conditioned C^1 -element matrices, and the measured $\sim 10^{-4}$ round-off floor of high-order Legendre–Ritz assembly for the free plate (non-monotone in

basis size beyond ~ 96 functions/axis).

References

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